



College Algebra: Systems of Linear Equations in Context — A Full Problem-Solving Session

College Algebra • Mathematics

COURSE-OBJ.MATH101.2

Translate a real-world scenario into a system of two linear equations, solve it using substitution or elimination, and correctly interpret what the solution means in context. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

PRINT • TEACH • REVISIT



Plan the session

Course and objective

College Algebra · Mathematics; COURSE-OBJ.MATH101.2 — Set up and solve systems of two linear equations in two variables using substitution and elimination, and interpret the solution in the context of a real problem.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both worked passages + vocabulary 15 min · guided systems organizer 18 min · partner check-and-discuss 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.



Before you begin

Key terms

system of equations — two or more equations sharing the same variables

substitution — solving one equation for a variable, then substituting it into the other

elimination — adding/subtracting equations to cancel a variable

break-even point — the input where two totals become equal

no solution — parallel lines; the system is never true at once

infinitely many solutions — the two equations describe the same line

Concept focus

A system of two linear equations in two variables has a solution wherever both equations are true at once — graphically, where the two lines cross. Substitution solves one equation for a variable and substitutes that expression into the other equation. Elimination adds or subtracts multiples of the equations to cancel a variable. Both methods always give the same answer; the choice is about which is faster for a given system. A system can have exactly one solution (lines cross once), no solution (parallel lines), or infinitely many solutions (the same line written twice) — checking the answer in both original equations catches arithmetic errors before they reach the final interpretation.

Model 1 — Solving by substitution

A gym charges a \$40 sign-up fee plus \$15/month ($y=15x+40$); a rival gym charges no sign-up fee but \$25/month ($y=25x$). Setting them equal: $15x+40=25x$, so $40=10x$, $x=4$. At $x=4$ months, both cost $y=\$100$ — the break-even point.

Model 2 — Solving by elimination

$x+y=12$ and $x-y=2$. Adding the equations cancels y : $2x=14$, so $x=7$; substituting back, $y=5$.
Check: $7+5=12$ ✓ and $7-5=2$ ✓.



WARM-UP

Warm-up

Answer using what you already know or recall from the previous session.

1. Write the equation for Peak Fitness's total cost after x months, and identify what each number in the equation represents.

2. Write the equation for Iron Works's total cost after x months, and explain why there is no constant term.



The Two Gyms Problem: Setting Up the System

Two gyms are competing for new members near a college campus. Peak Fitness charges a one-time \$40 enrollment fee plus \$15 per month. Iron Works charges no enrollment fee but \$25 per month. A student, Yuki, wants to know exactly when — if ever — the two gyms would cost the same total amount, so she can choose based on how long she plans to stay enrolled, not just the sticker price.

To answer this precisely, Yuki writes an equation for each gym's total cost after x months. For Peak Fitness, the total cost starts at the \$40 fee and adds \$15 for every month: $y = 15x + 40$. For Iron Works, the total cost starts at \$0 and adds \$25 for every month: $y = 25x$. Together, these two equations form a system — and the question "when do the costs match?" is really asking: for what value of x are both equations true at the same time?

Yuki solves the system using substitution, since both equations are already solved for y . Setting the two expressions equal: $15x + 40 = 25x$. Subtracting $15x$ from both sides: $40 = 10x$, so $x = 4$. At exactly 4 months, both gyms cost the same total amount: substituting $x=4$ into either equation gives $y = \$100$. Before month 4, Iron Works is actually cheaper, since Peak Fitness is still paying off its \$40 fee; after month 4, Peak Fitness becomes the cheaper option and stays cheaper for every month beyond that, since its rate per month is lower.



The Two Gyms Problem: What the Solution Means

A common mistake is stopping at " $x=4$ " without saying what it means. In this context, $x = 4$ means "after 4 months of membership," and $y = 100$ means "both gyms will have cost exactly \$100 in total by that point." The solution to a system is not just a pair of numbers — it is the answer to the real question being asked, expressed using the correct units and phrased in a full sentence.

Yuki double-checks her answer by substituting $x=4$ back into both original equations separately, rather than trusting the algebra alone. Peak Fitness: $y = 15(4) + 40 = 60 + 40 = 100$. Iron Works: $y = 25(4) = 100$. Both equations give $y=100$ at $x=4$, confirming the solution is correct in both equations, not just the one she solved for last.

Yuki then uses the solution to make her actual decision: she plans to be at the college for at least a full academic year, well beyond 4 months, so she chooses Peak Fitness, since it becomes and stays the cheaper option for any membership longer than 4 months. If she had only planned to stay for 2 months, Iron Works would have been the better choice instead — the same system of equations supports either decision, depending on the real constraint (how long she plans to stay) that the algebra alone cannot answer.



Guided practice · Reasoning organizer

Work through this organizer together before independent practice.

Given / Evidence

State the given information, quote, or data.

Method / Reasoning

Explain the method or reasoning in your own words.

Conclusion

State the answer or claim it supports.

1. Solve the system $15x+40=25x$ for x , showing each step.

2. Using $x=4$, find y for each gym's equation separately, and explain why both should give the same result.

3. Explain, in a full sentence using correct units, what " $x=4$, $y=100$ " means in the context of this problem.

4. Solve this new system using elimination and show your work: $x+y=12$ and $x-y=2$.

Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, check the elimination answer ($x=7$, $y=5$) in both original equations and explain why checking both — not just one — matters.

2. With a partner, discuss: for months 1 through 3, which gym is cheaper, and how does this relate to the $x=4$ solution?

3. Discuss: why is $x=4$ called a "break-even point," and what would change about Yuki's decision if she planned to stay for exactly 4 months?

Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Set up (but do not solve) a system of equations for two phone plans: Plan A costs \$20/month with no activation fee; Plan B costs \$10/month plus a \$60 activation fee.

2. Solve the phone plan system from the previous question using substitution, showing every step.

3. Interpret the phone plan solution in a full sentence with correct units.



Independent practice

Work independently. Show your work or cite evidence in every answer.

4. Solve this system using elimination and show your work: $2x+3y=16$ and $2x-y=4$.

5. Explain what it would mean, graphically, if a system of two linear equations had no solution, and give an example of when this could happen in a real cost-comparison scenario.

6. Explain what it would mean if a system of two linear equations had infinitely many solutions, and why this would be an unusual result for two different pricing plans.



Small-group extension work

Use for review or additional application practice.

1. A classmate solves $15x + 40 = 25x$ by writing " $x = 40/10 = 4$ " without showing the subtraction step. Explain what step they skipped and why showing it matters.

2. Create your own original two-plan cost-comparison scenario (e.g., ride-share apps, streaming services) and write the system of two equations that models it.



Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Solve the system you created in the previous question, showing your full method (substitution or elimination).

2. Interpret your solution from the previous question in a full sentence with correct units, matching the format used in the gym example.

Assessment

Complete independently to show what you learned today.

1. Self-assessment: which method (substitution or elimination) do you personally find faster, and describe one type of system where you would choose the other method instead.



Assessment · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.



Answer key

1. $y = 15x + 40$; 15 is the cost per month, x is the number of months, and 40 is the one-time enrollment fee (the constant/ y -intercept).
2. $y = 25x$; there is no constant term because Iron Works has no enrollment fee, so the total cost starts at \$0 when $x=0$.
3. Subtract $15x$ from both sides: $40=10x$. Divide both sides by 10: $x=4$.
4. Peak Fitness: $y=15(4)+40=100$. Iron Works: $y=25(4)=100$. Both give $y=100$ because $x=4$ is the point where the system is satisfied — both equations are true at once by definition of a system's solution.
5. After 4 months of membership, both Peak Fitness and Iron Works will have cost a total of \$100.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. Adding the equations: $2x=14$, so $x=7$. Substituting into $x+y=12$: $7+y=12$, so $y=5$. Solution: $x=7$, $y=5$.

7. $x+y=12 \rightarrow 7+5=12$ ✓; $x-y=2 \rightarrow 7-5=2$ ✓. Checking both matters because an arithmetic slip in one equation might still satisfy the other equation by coincidence, so only checking both confirms the solution is genuinely correct.

8. For months 1-3, Peak Fitness is more expensive because the \$40 fee has not yet been "paid off" by its lower monthly rate; Iron Works is cheaper until the two lines cross at $x=4$, after which Peak Fitness becomes and stays cheaper.

9. It is called a break-even point because the two costs are exactly equal there — neither gym is more or less expensive. If Yuki planned to stay for exactly 4 months, both gyms would cost the same total (\$100), so the choice would not matter financially and she could decide based on other factors (location, equipment, etc.).

10. $y=20x$ for Plan A; $y=10x+60$ for Plan B, where x is the number of months and y is the total cost.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. $20x=10x+60$. Subtract $10x$ from both sides: $10x=60$. Divide by 10: $x=6$. Substituting into $y=20x$: $y=120$. Solution: $x=6$, $y=120$.

12. After 6 months, both phone plans will have cost a total of \$120.

13. Subtracting the equations: $(2x+3y)-(2x-y)=16-4$, so $4y=12$, $y=3$. Substituting into $2x-y=4$: $2x-3=4$, so $2x=7$, $x=3.5$.

14. No solution means the two lines are parallel and never cross — the two costs are never exactly equal at any x . This could happen if two plans had the same monthly rate but different starting fees; the gap between their total costs would stay constant and never reach zero.

15. Infinitely many solutions means the two equations describe the same line; every x -value would be a solution. This would be unusual for pricing plans because it would mean two "different" plans actually have identical cost structures (same rate and same starting fee) — effectively the same plan.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. They skipped subtracting $15x$ from both sides (or an equivalent step) before dividing; showing it matters because it demonstrates the equation was solved correctly rather than the answer being guessed or memorized, and it helps catch errors in later, more complex systems.

17. Answers vary; must include two distinct linear equations in the same two variables (total cost y and units x , e.g., months or rides), each with realistic-looking constants matching a described real scenario.

18. Answers vary; work must correctly solve the student's own system and arrive at a solution consistent with their two equations.

19. Answers vary; must state the x -value with correct units (e.g., "after 5 rides") and the y -value with correct units (e.g., "both apps cost \$45"), matching the interpretation style modeled in the lesson.

20. Answers vary; must name a genuine preference and describe a specific system type (e.g., "when both equations are already solved for y , substitution is faster; when the coefficients line up for easy cancellation, I'd use elimination instead").

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment • accessed September 18, 2026

COURSE-OBJ.MATH101.2 — Set up and solve systems of two linear equations in two variables using substitution and elimination, and interpret the solution in the context of a real problem. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

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